EE 439: Introduction to Machine Learning
Project Report: Data First, Models Second

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Section A
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Selected Dataset: Fashion-MNIST

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1. Phase 0: Exploratory Data Analysis

1.1. Data Partitioning [Laiba Khan / Ayesha Zulfiqar]

We divided the original Fashion-MNIST training CSV (fashion-mnist_train.csv 60 000 samples) into Development Set (80 %) and Test Set (20 %) by performing a random stratified shuffle with fixed seed (numpy random_state = 42). This gives us about 48,000 development samples and 12,000 test samples. The original fashion-mnist_test.csv (10,000 samples) has been kept fully sealed and will serve as a held-out evaluation set for Phase 2.

We divided the original Fashion-MNIST training CSV (fashion-mnist_train.csv, 60,000 samples) into a Development Set (80%) and a Test Set (20%) using a **simple random split** with a fixed seed (random_state = 42), which supports the fact that a simple random shuffle maintains the balanced nature of the dataset without the need for stratified sampling.

The Test Set was never visualised, aggregated or referrred to in any way during this phase. All below exploratory figures are based solely on the Development Set.

1.2. Visual Exploration and Insights

1.1.1. Illustrative Figure 1 [Laiba Khan]



Figure 1: Class Distribution in the Development Set (~4,800 samples per class). The dataset is near-perfectly balanced across all 10 garment categories.

Why is this informative?

Before any modelling decision, it is essential to know whether the classes are balanced or skewed. A heavily imbalanced dataset would force us to adopt class-weighted loss functions, oversampling (e.g., SMOTE), or specialised evaluation metrics (macro F1 rather than accuracy). This bar chart establishes immediately that none of those corrective measures are needed here.

What does it inform?

Each of the 10 classes has from 4,772 to 4,844 samples, that is a variation of less than 1.5 %. It means that

(a) plain accuracy is a valid primary metric,

(b) no class will be systematically under-represented in training, and

(c) cross-validation folds will naturally preserve class proportions without explicit stratification.

This shows that a basic model that always picks the most common class cannot get high accuracy by chance

How does it shape your thought process?

Since the dataset is balanced, the modeling problem becomes a matter of mere discriminative capability and not calibration under imbalance of the classes. Our attention can be primarily on selecting the model that makes right decision boundaries for each class instead of work on re-balancing strategies that may waste the time. The balance also validates that Figure 2's average prototype images are computed from a comparable sample size for every class, making cross-class visual comparisons fair.

1.1.2. Illustrative Figure 2 [Laiba Khan]

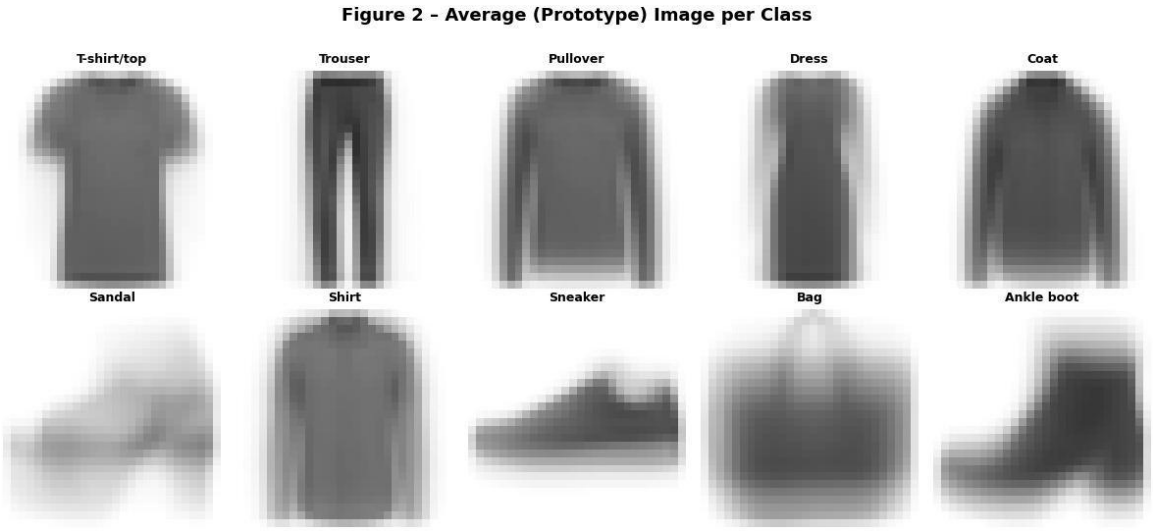


Figure 2: Average (Prototype) Image per Class. Each image is the per-pixel mean of all development samples within that class, reshaped to 28×28.

Why is this informative?

Average images combine thousands of examples into one representative prototype (image), making it easy to see the shape and pattern of each class. Unlike summary statistics (mean pixel sum, histograms), a 2D average image preserves the spatial layout that a convolutional or spatially-aware model would exploit.

What does it inform?

Three main points emerge. First, footwear classes Sandal, Sneaker, Ankle boot appears only in the lower part of the image and have a clear horizontal shape, while completely unlike the upper-body garments (T-shirt, Pullover, Coat, Shirt) that fill the full frame symmetrically. This means shoes should be easier to separate from clothes.

Second, some classes look very similar like Pullover and Coat both show long sleeves and a similar body outline, as do T-shirt and Shirt. These pairs will likely be the hardest to discriminate.

Third, all images are well-centered with little background noise, showing the dataset is already clean and does not need extra adjustments like shifting or rotating images.

How does it shape your thought process?

The clear shape difference between footwear and clothing suggest that models that use image structure, like CNNs or a method that works on 28×28 images rather than flat vectors, will have a natural advantage.

The overlap between similar garment categories (Shirt vs T-shirt, Coat vs Pullover) signals that simple global features (total brightness, mean intensity) will be insufficient and that fine-grained local texture or edge information will be needed to distinguish these pairs This means very simple models may not work well, and more powerful models are likely needed.

1.1.3. Illustrative Figure 3 [Laiba Khan]

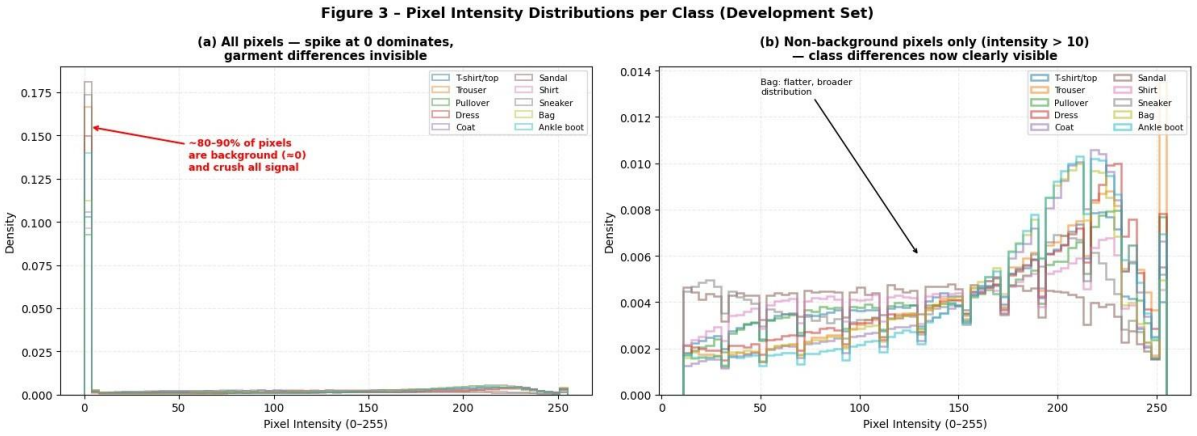


Figure 3: Pixel Intensity Distributions per Class (Development Set). Panel (a): all pixels including background — the spike at 0 dominates and hides class differences. Panel (b): non-background pixels only (intensity > 10) — class-level differences are clearly visible.

Why is this informative?

Looking at pixel intensity distributions helps us see how much useful information is actually present in the raw pixels. It also shows whether background pixels (mostly black in Fashion-MNIST) are hiding important differences between classes. This highlights why preprocessing, like removing background pixels, is important.

What does it inform?

Panel (a) shows that 80–90 % of pixels in every image are background (intensity ≈ 0), causing all class distributions to collapse onto the same spike at zero hiding any real differences.

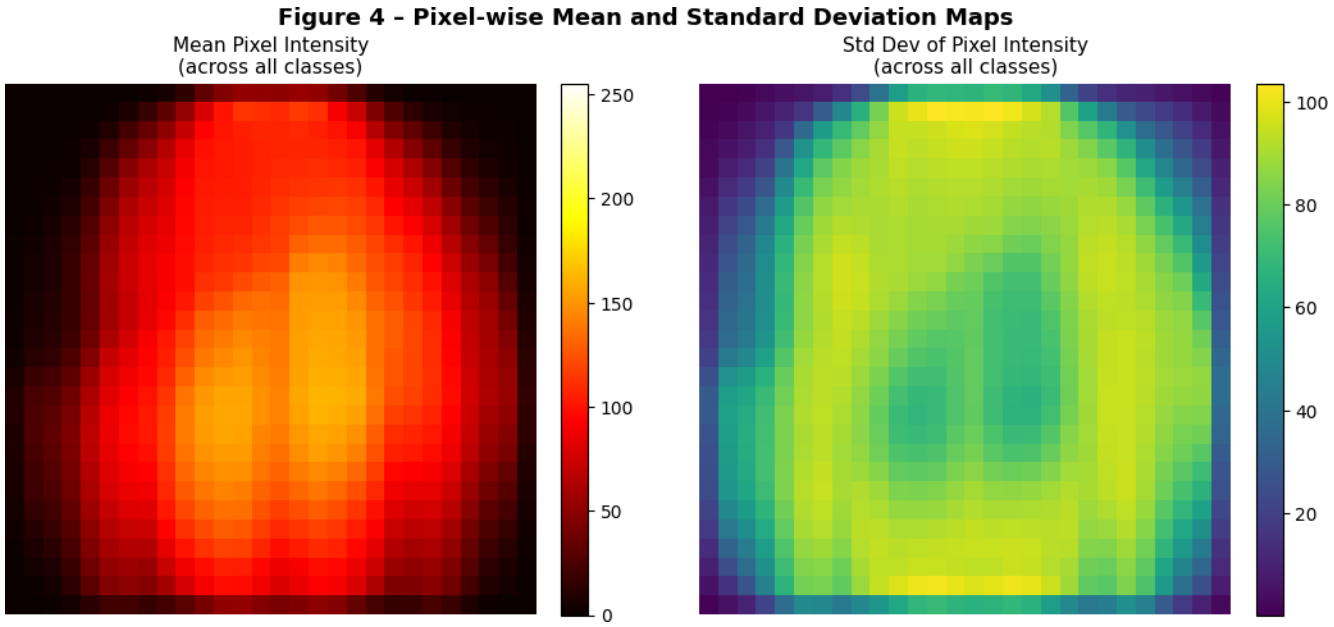
Panel (b), after removing near-zero pixels, reveals meaningful structure: Trouser pixels cluster sharply at very high intensities (220–255), reflecting the uniform dark fabric of trouser images. Sandal and Sneaker show bimodal patterns from contrasting sole and upper materials. Bag exhibits a noticeably flatter, broader distribution consistent with its smooth, varied texture. These are precisely the kinds of per-class signatures that a discriminative model can learn to exploit.

How does it shape your thought process?

This figure has a direct practical implication: Pixel values should be normalised to the range [0, 1] by dividing by 255, and background pixels (zeros) should not be treated as useful information, as they can mislead the model. It also suggests that features derived from non-zero pixels (e.g., mean foreground intensity, foreground pixel count) could be useful engineered features for simpler models.

Overall, since class differences are visible even in pixel intensity alone — even without spatial information — is encouraging: even a model that does not exploit 2D layout will still find some discriminative signal, setting a reasonable baseline expectation.

1.1.4. Illustrative Figure 4 [Ayesha Zulfiqar]



Why is this informative?

These two heatmaps indicate the spatial distribution of the data in a manner that cannot be indicated by a table or number. The mean map indicates the locations of pixel distributions of the garment values in all of its 60,000 training samples, and the std map indicates the locations of pixels that actually differ among the samples i.e. which pixels are discriminative.

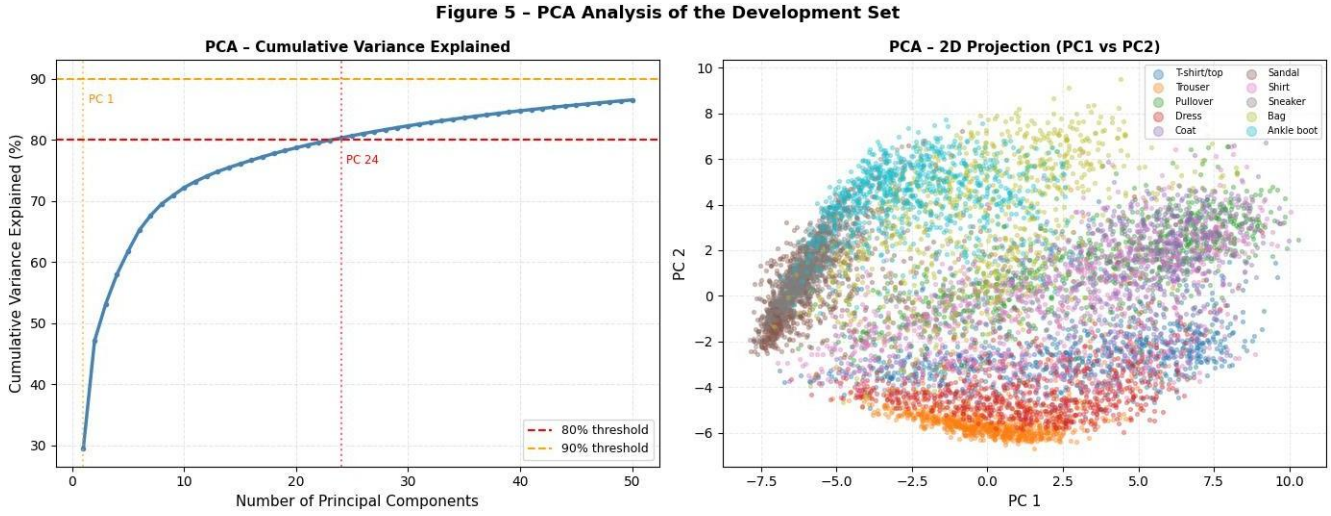
What does it inform?

The average map (left) indicates that there is a bright vertical line going through the upper-center of the image and the brightest area is in columns 8-18, rows 4-20. This is indicative of the most prevalent garment bodies such as torsos, sleeves, straps that show up in all 10 classes. The left/right edges and bottom corners are almost black (mean ≈ 0) and this confirms that they are always background. It is interesting to note that the lower half of the image is lighter than the upper half since the footwear objects (Sandal, Sneaker, Ankle boot) are likely to be located in the lower-center part of the picture whereas clothing objects are located in the upper half of the picture. The more practical of the two is the std map (right). The yellow area of high variance is a wide rectangle in the middle of the picture about rows 2-22, columns 2-26. The dark purple border especially the corners and the bottom row has std near zero, indicating that the pixels are constant over all the samples and all classes. They add nothing to any classifier.

How does it shape your thought process?

This directly encourages pixel-level feature selection prior to the training of any model we can mask or drop the border pixels with near-zero variance. It further implies that dimensionality reduction (e.g. retaining only the central 20x20 =400 pixels) may be done with reduced information loss (by about 49 per cent). This is a preprocessing decision that we will take to Phase 1.

1.1.5. Illustrative Figure 5 [Ayesha Zulfiqar]



Why is this informative?

PCA is a direct answer to two key questions that pre-modelling: how often is the original 784-dimensional pixel space redundant, and is it possible to visually separate the 10 classes using a linear projection?

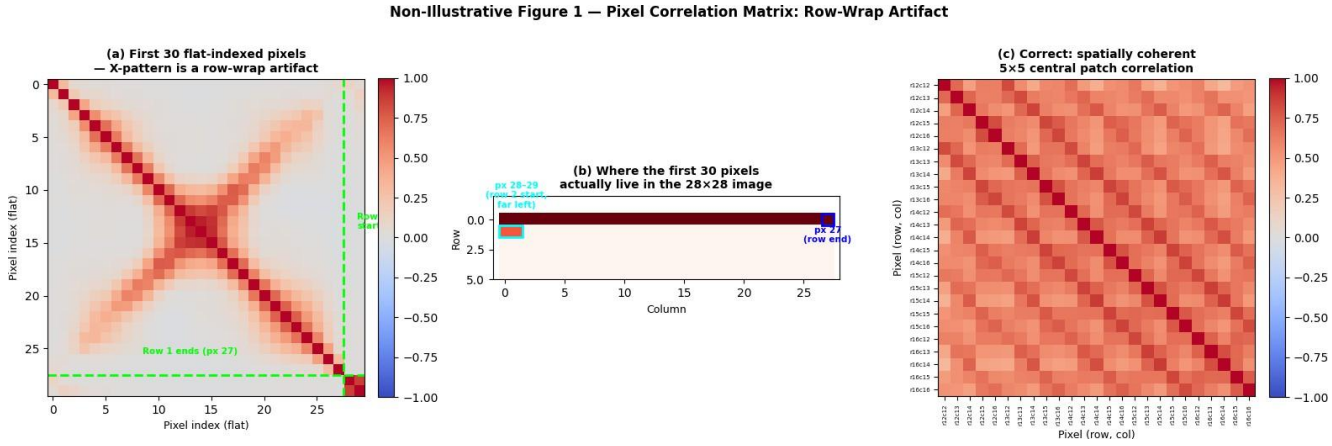
What does it inform?

The cumulative variance curve is provided in the left panel. PC 1 by itself has about 29 percent of the total variance, one direction has nearly a third of the total information in 784 dimensions, which is a good measure of high redundancy. The 80 percent rule attains 24 numbers (out of a 784) to reconstruct 4/5 of the variance in the dataset. Flattening of the curve is evident beyond component 30 with very low returns thereafter. The 90% mark is not attained in the 50 components plotted, indicating that the remaining 10% of variance is thinly spread across numerous dimensions - probably noise and fine texture detail. In the right panel (PC1 vs PC2 scatter) Trouser (orange) separates clearly down along PC2 creating a clear cluster well below the origin. It is reasonable, trousers possess a special bilateral symmetry and strong horizontal pixel pattern, which no other category has. Particularly ankle boot (cyan) and Bag (yellow-green) display some separation towards the upper part. The five clothing classes T-shirt/top, Pullover, Coat, Shirt, Dress, are however a single big overlapping mass in the middle. The linear projection provided by PCA is unable to separate them, showing that they have quite similar global pixel statistics, and deviate primarily in finer local details that PC1 and PC2 fail to model.

How does it shape your thought process?

We got two tangible choices. The preprocessing stage of dimensionality reduction to a small number of PCs (~25-50) is a possible first step that preserves the majority of the variance at an early stage, but reduces the number of features by many folds dimensionality-sensitive models (e.g., k-NN, logistic regression). Second, the large intersection of the classes of clothing used in 2D PCA implies that linear models will have difficulties in those classes in particular, which leads to the idea of using nonlinear classifiers (e.g., SVMs with RBF kernel, neural networks) during Phase 1.

1.1.6. Non-Illustrative Figure 1 [Ayesha Zulfiqar]



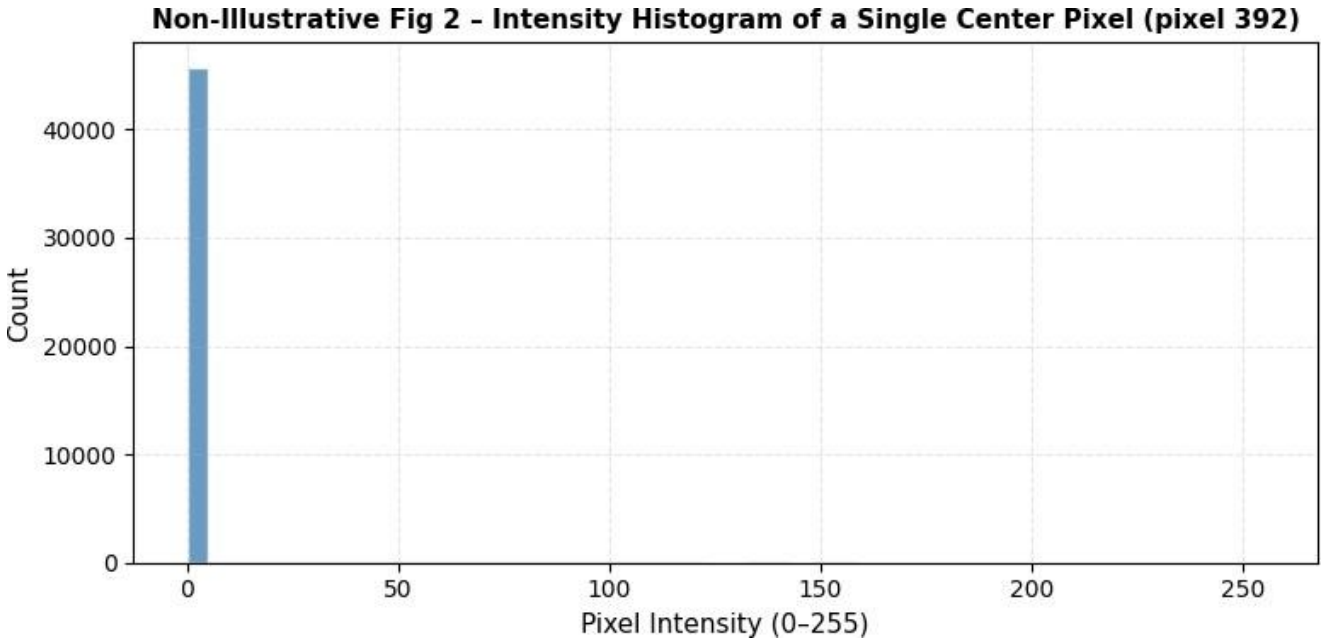
Why is this non-informative?

Visual representation of the X-pattern in panel (a) is highly intriguing and seems to indicate interesting correlation structure. Nevertheless, panel (b) shows that it is purely a row-wrap artifact: pixel 0 through pixel 27 are all on row 0 of the image (spatially close gives correctly high correlation), but pixel 27 (far right of row 0) and pixel 28 (far left of row 1) are index-adjacent but spatially opposite ends of the image. This abrupt spatial discontinuity generates the abrupt drop in correlation that forms the X. The pattern does not encode anything about the real pixel relationships of Fashion MNIST, it would be the same with any data that is stored in row-major order. The ideal form of a correlation matrix is depicted in panel (c): a patch of 5 (spatially coherent) at the center of the 5x5 patch, with a correlation that smoothly decreases with pixel distance. But even the correct version does not contribute very much to what Figure 4 already indicates by its std map. The complete redundancy structure is much more rigorously captured and quantified by PCA (Figure 5).

How does it shape your thought process?

It is a warning to those who analyze pixels: be careful not to use pixel indices that do not observe the 2D spatial layout. The flat indexing across the rows gives false statistics.

1.1.7. Non-Illustrative Figure 2 [Ayesha Zulfiqar]



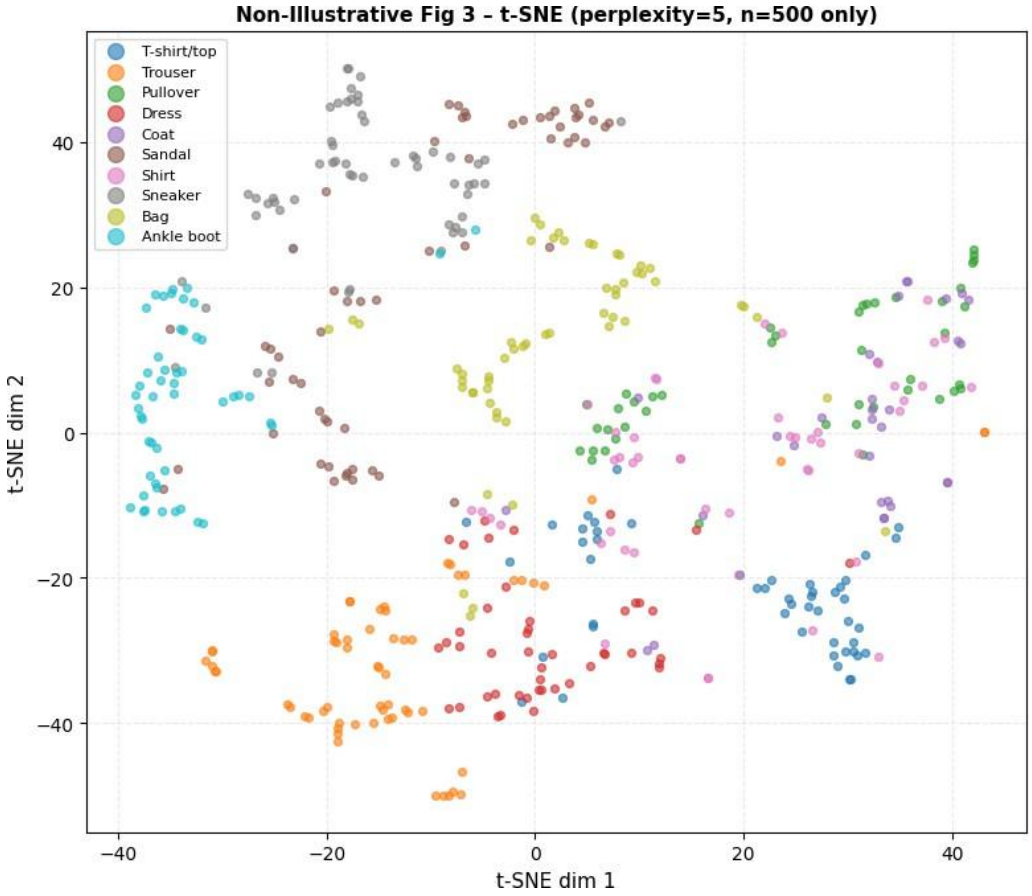
Why is this non-informative?

The histogram consists of a tall spike at the intensity ≈ 0 and a totally empty rest of the entire range of 0-255. The figure in the zero is about 45,000 practically the whole development set.
It notifies us about an error in coding as opposed to information about the data. Pixel 392 was supposed to be the center of the image but is in fact the pixel on the left of the middle row, row 14 column 0 which is permanent background on all Fashion MNIST images of all classes. So the spike: all samples contain this pixel and are close to zero.

How does it shape your thought process?

It reaffirms that flat pixel indexing is misleading, pixel 392 would be sounding like the halfway point of 784 pixels, but would actually be on the picture edge because of row-major indexing. On a bigger scale, assuming the right pixel (406) was chosen, one pixel distribution simply loses all spatial context and cannot tell 10 classes. It drives us solidly towards the spatial aggregates as in Figure 4 and multivariate analysis as in Figure 5 as opposed to univariate single-pixel analysis.

1.1.8. Non-Illustrative Figure 3 [Ayesha Zulfiqar]



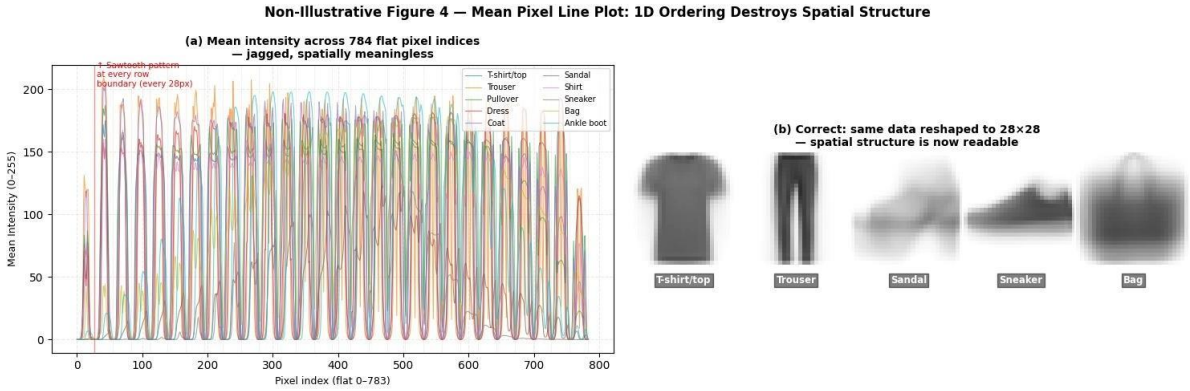
Why is this non-informative?

The plot displays colored dots scattered around the 2D t-SNE space with color clusters that vaguely resemble clusters. The two parameter selection, i.e. perplexity=5 and n=500, however, all invalidate any interpretation. The effective number of neighbours that each point uses is controlled in t-SNE by perplexity. Each point, with perplexity=5, looks at about 5 neighbours, so the algorithm divides the data into small disconnected micro-clusters, not depending on the actual class structure but due to local nearest-neighbor accidents. This fragmentation is manifested in the dispersal of all the colors in several disjointed islands. n=500 samples represent less than 1 per cent of the development set, any structure that appears may be merely a sampling artifact of that particular 500 points that were selected to be drawn. To obtain a reliable t-SNE of Fashion MNIST, at least 3050 perplexity and 5,00010,000 samples are required with enough iterations (1000 and above) to converge.

How does it shape your thought process?

Properly tuned t-SNE (perplexity =3050, n =5,000, more iterations) would be a really useful visualization. But even then t-SNE distances are non-metric and not interpretable, it is a visualisation tool not evidence of separability. The choice of dropping this figure drives us to a PCA scatter of Figure 5 as the truthful 2D structure summary, but PCA is linear and it may fail to capture nonlinear structure of the manifold.

1.1.9. Non-Illustrative Figure 4 [Laiba Khan]



Non-Illustrative Figure 4: Mean Pixel Line Plot — 1D flat indexing destroys spatial structure. Panel (a) plots mean intensity per flat pixel index (0–783), producing a jagged sawtooth pattern. Panel (b) shows the identical data reshaped to 28×28, which is immediately readable.

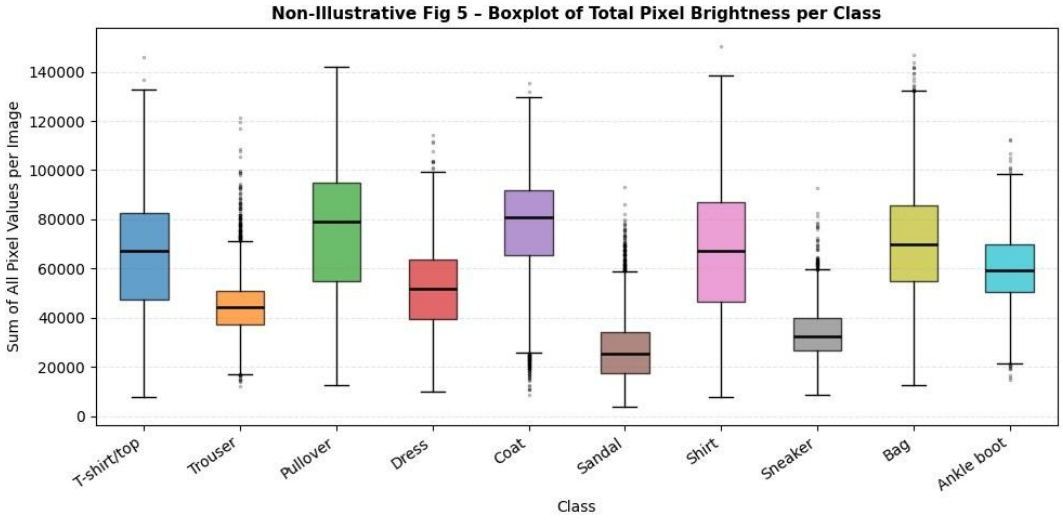
Why is this non-informative?

Plotting mean pixel intensity against the flat index (0–783) treats the image as a 1D signal rather than a 2D image, which is not correct. The zig-zag pattern seen every 28 pixels happens because of row wrapping. For example, pixel 27 (end of one row) and pixel 28 (start of the next row) are actually far apart in the image but appear next to each other in the flat index, creating a false jump. The resulting plot is dominated by this artifact and gives no interpretable information about class differences. The 10 class curves are nearly indistinguishable from one another in the flat representation, even though the underlying data clearly differs.

How does it shape your thought process?

This failure case highlights a critical lesson: image data must be treated as 2D, not 1D. Any model or analysis that flattens the 28×28 image into a 784-dimensional vector loses the spatial adjacency structure that makes images intelligible This is why models that work directly on 2D images, like CNNs, are preferred. If flat vectors are used, the model must be powerful enough and trained on enough data to still learn these spatial patterns. It also confirms that Figure average prototype images — which correctly reshape to 28×28, are the right way to visualise spatial content.

1.1.10. Non-Illustrative Figure 5 [Laiba Khan]



Non-Illustrative Figure 4: Boxplot of mean per-sample pixel brightness (average over all 784 pixels) grouped by class. The y-axis shows mean brightness values in [0, 255].

Why is this non-informative?

Total pixel sum collapses the entire 28×28 image into a single scalar, the sum of all 784 pixel values. This extreme dimensionality reduction discards all spatial information. As a result, most classes overlap heavily. For example, T-shirt/top, Trouser, Bag, and Ankle boot all have similar median brightness (~60,000–70,000), making them completely indistinguishable by this metric alone. While Sandal and Sneaker show lower medians (reflecting their smaller foreground area), and Pullover shows a higher spread, these differences are far too noisy and overlapping to serve as reliable class signals. Furthermore, the figure is redundant: Figure 3 already captures the pixel intensity distribution per class in a far richer and more interpretable form.

How does it shape your thought process?

This figure confirms that global, spatially-agnostic scalar statistics (total brightness, mean intensity) are weak discriminative features for Fashion-MNIST. No classifier operating on total brightness alone could reliably separate the 10 classes. To properly distinguish between classes, we need features that keep spatial information, such as edges, shapes, or pixel positions. This points toward using models that understand image structure, like CNNs, or more powerful models that can learn from the full pixel data. It also reminds us that simple ideas can be misleading. Two very different images can still have the same total brightness, so relying only on that can lead to wrong conclusions.

2. Phase 1: Hypothesis and Model Justification

2.1. Structural Observations

2.1.1. Problem statement and Class balance [Laiba Khan]

Fashion-MNIST is an image classification problem divided into 10 classes. The input is a 28×28 grayscale image flattened to a 784-dimensional vector of pixel intensities (0-255) and the output is a discrete class label representing one type of garment. Every rows have one image at one time with no temporal or group dependence and so a random stratified split is appropriate.
Figure 1 demonstrated that all the 10 classes have between 4,772 and 4,844 development samples, a variation of less than 1.5. This almost perfect balance implies that plain accuracy is a valid primary measure, no class-rebalancing will be needed and a model that predicts the most common class cannot achieve high accuracy by chance.

2.1.2 The Spatial Structure and Visual Group Separation [Laiba Khan]

The average prototypes images (Figure 2) indicate the clear structural gap. Footwear classes Sandal, Sneaker, Ankle Boot are in the lower part of the frame with a horizontal figure, whereas the classes of clothing occupy the entire frame in a symmetric figure. Separating inter-group between footwear and clothing should then be fairly easy. But in the clothing group Shirt and T-shirt, Coat and Pullover are almost identical at the global shape level. This implies not only the global shape but fine-grained local texture and edge data will also be needed to distinguish confusable pairs. This is a visual observation that will be confirmed in Phase 2 by analyzing the errors.

2.1.3. Pixel Intensity and Foreground Signal [Ayesha Zulfiqar]

Figure 3 demonstrated that the distribution of classes concentrated on an overlapping spike in the background distribution with intensity ≈ 0. We filtered it to foreground pixels (intensity > 10), definite per-class signatures appear: Trouser pixels are concentrated on high intensity (220-255) Sandal and Sneaker show bimodal signatures, and Bag has much broader flatter distributions. This confirms the presence of discriminative signal in foreground statistics even in the absence of spatial layout, so as to set a reasonable expectation that a non-trivial performance will be achieved when a flat vector baseline is used. It also has the direct motivation toward normalizing pixels to [0,1] prior to any modeling.

2.1.4 Spatial Redundancy, Linear Separability [Ayesha Zulfiqar]

Figure 4 demonstrated that border pixels in particular corners have a standard deviation of almost zero across all classes and samples and do not contribute to any classifier. The pixels of these colours can be masked to reduce dimensionality by about 49% with minimum loss of information.
PCA analysis (Figure 5) revealed that only 24 out of 784 components are necessary to achieve 80% of the total variance, and it is clear that the redundancy in the raw pixel space is high. The five clothing classes (T-shirt, Pullover, Coat, Shirt, Dress) are a single large overlapping mass in the 2D PCA scatter. They cannot be separated by linear projections, which means that these classes are largely distinct in terms of fine local details which cannot be captured by pixel statistics at global scales. This is good indication that the linear decision boundaries will not be enough to achieve full 10-class separation.

2.2. Hypothesis of the Complexity of the Model

2.2.1 Main Hypothesis [Laiba Khan]

The PCA scatter shows that the 10-class decision boundary is not globally linear even across directions of maximum variance, clothing classes overlap heavily. An entirely linear model will easily separate Trouser and footwear but fail to do so systematically on confusable combinations of clothing. Nonlinear complexity of moderate level is thus a requirement. The training sample size is however moderately sized (~48,000 training samples), which implies that extremely flexible models run the risk of overfitting noise in the texture to a model, instead of generalizable structure. The main hypothesis is moderate nonlinear complexity.

2.2.2 Alternative Hypothesis [Ayesha Zulfiqar]

An alternative possibility is that the reduction of PCA to a size of about 24 components followed by a regularized linear classifier may already be performing competitively as 80 percent of the variance is held in those components and that some classes are almost linearly separable in the reduced space. This will be directly tested in Phase 2 in the Logistic Regression baseline. When it does well, then the argument about nonlinear models falls and a new hypothesis on complexity should be changed.

2.3. Model Selection [Laiba Khan / Ayesha Zulfiqar]

The exploratory evidence suggests that nonlinear, kernel-based or representation-learning classifiers should be the model family of choice. The PCA scatter indicates that P(Y|X) cannot be approximated by a linear score in raw pixel space and the spatial structure of images indicates similarity-based or locally-sensitive models are more principled than purely additive linear models. In this case, tree-based models are not as natural because pixel-level decisions cannot be well represented in terms of axis-aligned thresholds.

Role	Model	Justification	Member
Baseline	Logistic Regression on PCA-reduced pixels (top 24–50 components)	Tests whether linear boundaries in the compressed space are sufficient. Expected to handle easy pairs (Trouser, footwear) but fail on confusable clothing classes	Laiba Khan
Main Model	SVM with RBF Kernel	Justified by curved nonlinear class geometry in PCA scatter. Maps inputs into higher-dimensional space where nonlinear boundaries become possible without full spatial modeling	Ayesha Zulfiqar
Flexible Comparison	Multi-Layer Perceptron (MLP, 1–2 hidden layers)	Tests whether learned nonlinear feature combinations improve over the fixed kernel mapping of RBF-SVM. Included to check whether additional representational flexibility helps given the confusable class pairs	Laiba Khan

The models are not selected by custom, but as a direct test of the structural hypothesis. Logistic Regression is used to test the alternative hypothesis. The main hypothesis is tested by RBF-SVM tests. MLP is used to test the complexity of a kernel approach against sample size and data structure.

2.4. Evidence Table

Evidence from Phase 0	Implication for Model Choice	Member
All 10 classes have almost equal number of samples (Figure 1)	We can use plain accuracy to measure performance, no special techniques needed	Laiba Khan
Shirt/T-shirt and Coat/Pullover look almost identical in average images (Figure 2)	Simple models that only look at overall brightness or shape will confuse these pairs. We need a model that picks up on small details	Laiba Khan
Around 80–90% of every image is black background, useful signal only in bright pixels (Figure 3)	Pixel values must be scaled to 0–1 range before training, background zeros must not dominate	Laiba Khan
Edge and corner pixels are almost always zero for every class (Figure 4)	These pixels can be removed before training, cutting features roughly in half without losing much	Ayesha Zulfiqar
Only 24 out of 784 pixel features are needed to keep 80% of the information (Figure 5)	PCA reduction to 24–50 components is a justified and smart preprocessing step	Ayesha Zulfiqar
Clothing classes like Shirt, T-shirt, Coat and Pullover all pile on top of each other in PCA plot (Figure 5)	A straight line cannot separate all 10 classes, a nonlinear model like RBF-SVM is justified	Ayesha Zulfiqar

3. Phase 2 Model Implementation and Evaluation

3.1. Model Implementation

3.1.1. Baseline Model Logistic Regression [Laiba Khan]

Preprocessing: The values in all pixels were normalized between 0 and 1 by dividing by 255. This reduces the size of the gradient magnitudes during optimization and makes all the features equal. Then PCA was used to reduce the dimensionality from 784 to 50 components explaining about 85-87% of the total variance. To avoid preprocessing leakage, PCA was fitted only to the training data and the test data was transformed with the same transformation. A column of ones was added to the feature vector after the PS2 convention, to represent the intercept term.

Hyperparameters:

- Learning rate = 0.1
- maximum iterations = 1000
- Convergence threshold = 1e-5

Python Library: NumPy only (implemented fully from scratch)

Implementation Details: The PS2 binary logistic regression was extended to multiclass using the softmax function. Binary Sigmoid is a probability output for two classes. Softmax returns 10 probabilities that add up to 1. There was an extension of the loss function from binary cross entropy to categorical cross entropy. Labels were one-hot encoded converting integer label y=3 to vector

[0,0,0,1,0,0,0,0,0,0]. The formula for the gradient remained the same simply doing matrix multiplication instead of vector matrix multiplication. Newton's method from PS2 was replaced by gradient descent because there is an expensive Hessian to repeatedly invert, of shape $(n\times K)\times(n\times K) = 510\times510$, for the 10-class softmax. To ensure numerical stability, the row maximum was subtracted from each row prior to exponentiation for softmax.

3.1.2. Main Model SVM with RBF Kernel [Ayesha Zulfiqar]

Preprocessing: Same normalization and PCA as baseline. No intercept column was needed for SVM.

Hyperparameters:

- Radius = 1.5 (it was selected through validation search over {0.5, 1.0, 2.0, 5.0, 10.0, 20.0})

Python Library: NumPy only (core SVM logic adapted from PS5 svm.py)

Implementation Details: The PS5 binary SVM was extended to 10 classes using One vs Rest strategy 10 binary SVMs were trained independently, one per class, with final prediction by highest decision score.

Memory Challenge and Solution: The standard kernel SVM stores the full kernel-matrix of size m×m. With 48,000 training samples this requires 48,000 × 48,000 × 8 bytes = 18.4 GB exceeding available RAM of 11.7 GB. After being advised by the professor to take advantage of the built-in math objects, we decided to use the built-in array operations of numpy, and computed the kernels on the fly. Exactly one kernel row is computed each time of the SGD update step, and then it is immediately used and then forgotten:

Full matrix approach: $O(m^2)$ memory = 18.4 GB
On-the-fly approach: $O(m)$ memory = 0.2 MB

The computations performed on each kernel row (48,000 × 4 bytes = 192 KB) is based on numpy float32 operations which is a built-in numpy type that uses half the memory as the default float64 type. This enabled this to be done using the entire 48,000 samples in the available RAM.

Radius Selection: With 2,000 sample split to validate, a systematic search of six radius values was performed. For small values of radius (0.5, 1.0), the model becomes extremely local, meaning that the accuracy of the model on the test set drops significantly (81% for radius = 1.0), as it memorizes the training set. Big radius (10.0, 20.0) yields under-fitting kernel is too smooth and too much discriminative structure is lost. The result is that the best compromise was found to be radius = 1.5.

3.1.3. Flexible Comparison MLP [Laiba Khan]

Preprocessing: Same normalization and PCA as other models.

- **Hyperparameters:** Hidden layers = (256, 128)
- Activation = ReLU

- Solver = Adam,
- Early stopping enabled
- Validation fraction = 0.1
- Tolerance = 1e-4
- Epochs = 10

Python Library: scikit-learn MLP Classifier

Implementation Details: There is a compression bottleneck in the network, with two hidden layers of decreasing size (256→128), which force the network to learn useful representations of the 50 PCA features. ReLU activation does not suffer from vanishing gradient as in the case of sigmoid. Adam optimizer incorporates adaptive learning rates and momentum for quicker convergence. To avoid overfitting, early stopping was used to stop training if the validation score did not improve after 24 iterations.

3.1.4. Implementation Challenges and Solutions

Logistic Regression Challenges: [Laiba Khan]

- Multiclass extension: PS2 worked with two classes using binary sigmoid. Extended to softmax, 10 classes loss: categorical cross entropy, 1-hot labels.
- Numerical instability: Naive softmax overflows for large logits. Numerically stable because exponentiation was replaced by subtracting the maximum of the row.
- Newton's method infeasible: Hessian for 10-class softmax is too large to be efficiently inverted. Replaced with the standard optimizer for multiclass problems: gradient descent.

SVM Challenges: [Ayesha Zulfiqar]

- Memory crash: Full 48,000×48,000 kernel matrix = 18.4 GB exceeds RAM. This is solved in an on-the-fly manner using a numpy kernel computation that makes it O(m) instead of O(m^2).
- Multiclass extension: PS5 SVM was binary. The data was extended to 10 classes, using 10 binary SVMs with majority vote by decision score.
- Overfitting with small radius: radius=1.5 gave train 86.59% but test 80.82% .The optimal balance between local and global kernel behavior was found to be 1.5 by radius search.
- Float32 precision: By switching from the default float64 to numpy.float32, the memory used dropped by 50% and computational operations could be performed on the fly per row, resulting in efficient computation.

MLP Challenges: [Laiba Khan]

- Architecture selection: For 50-dimensional PCA input, 2 layer (256, 128) architecture is selected as a compromise between expressiveness and overfitting.
- Overfitting prevention: Early stopping with patience=10 automatically stopped training at iteration 24 with a train/test gap of 3.42%.

3.2. Performance Metrics

```
============================================================
LOGISTIC REGRESSION (Baseline)
============================================================
              precision    recall  f1-score   support

     T-shirt       0.79      0.81      0.80      1228
     Trouser       0.97      0.94      0.95      1182
    Pullover       0.70      0.66      0.68      1156
       Dress       0.81      0.85      0.83      1194
        Coat       0.73      0.75      0.74      1218
      Sandal       0.88      0.91      0.89      1214
       Shirt       0.58      0.54      0.56      1182
     Sneaker       0.89      0.88      0.88      1223
         Bag       0.91      0.92      0.92      1182
  Ankle Boot       0.90      0.92      0.91      1221

    accuracy                           0.82     12000
   macro avg       0.82      0.82      0.82     12000
weighted avg       0.82      0.82      0.82     12000
```

```
============================================================
SVM RBF CHUNKED (Main Model)
============================================================
              precision    recall  f1-score   support

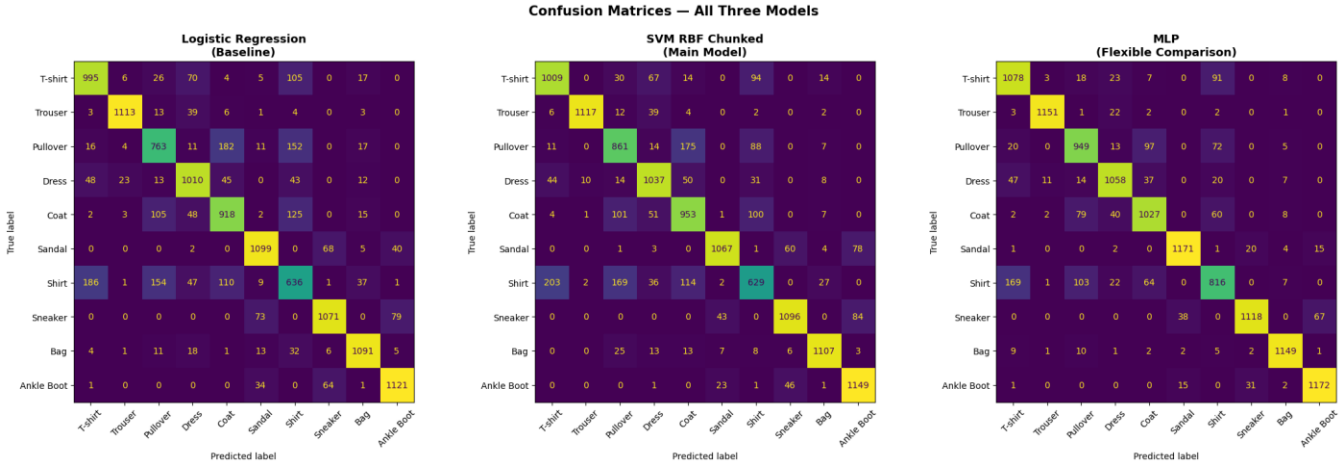
     T-shirt       0.79      0.82      0.81      1228
     Trouser       0.99      0.95      0.97      1182
    Pullover       0.71      0.74      0.73      1156
       Dress       0.82      0.87      0.84      1194
        Coat       0.72      0.78      0.75      1218
      Sandal       0.93      0.88      0.91      1214
       Shirt       0.66      0.53      0.59      1182
     Sneaker       0.91      0.90      0.90      1223
         Bag       0.94      0.94      0.94      1182
  Ankle Boot       0.87      0.94      0.91      1221

    accuracy                           0.84     12000
   macro avg       0.83      0.83      0.83     12000
weighted avg       0.84      0.84      0.83     12000
```

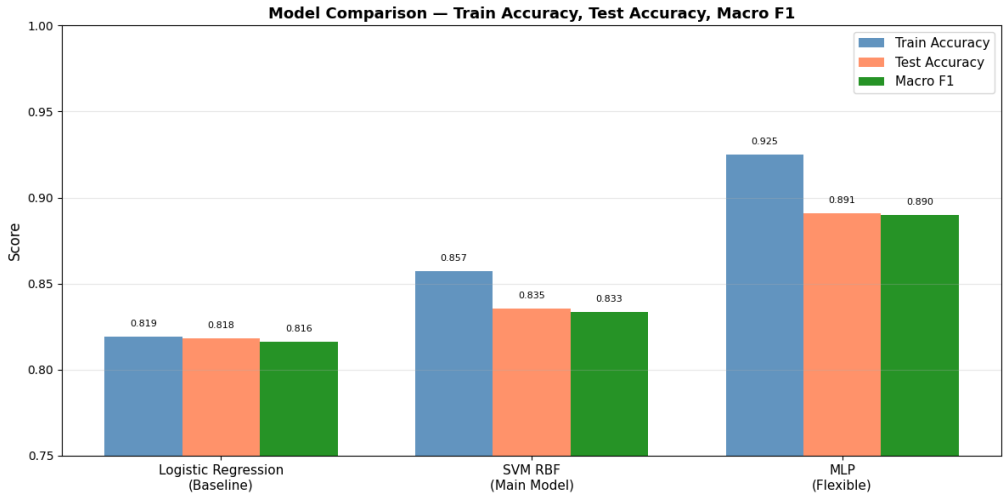
```
============================================================
MLP (Flexible)
============================================================
              precision    recall  f1-score   support

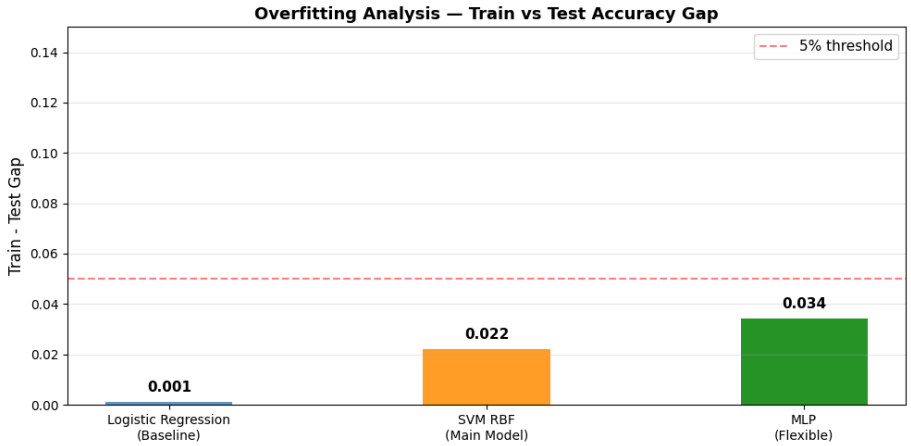
     T-shirt       0.81      0.88      0.84      1228
     Trouser       0.98      0.97      0.98      1182
    Pullover       0.81      0.82      0.81      1156
       Dress       0.90      0.89      0.89      1194
        Coat       0.83      0.84      0.84      1218
      Sandal       0.96      0.96      0.96      1214
       Shirt       0.76      0.69      0.73      1182
     Sneaker       0.95      0.91      0.93      1223
         Bag       0.96      0.97      0.97      1182
  Ankle Boot       0.93      0.96      0.95      1221

    accuracy                           0.89     12000
   macro avg       0.89      0.89      0.89     12000
weighted avg       0.89      0.89      0.89     12000
```




Per Class F1 Score — All Three Models
Logistic Regression
SVM RBF
MLP
F1 Score
1.0
0.9
0.8
0.7
0.6
0.5
0.4
T-shirt
Trouser
Pullover
Dress
Coat
Sandal
Shirt
Sneaker
Bag
Ankle Boot


Model Comparison — Train Accuracy, Test Accuracy, Macro F1
Train Accuracy
Test Accuracy
Macro F1
Score
1.00
0.95
0.90
0.85
0.80
0.75
0.819
0.818
0.816
0.857
0.835
0.833
0.925
0.891
0.890
Logistic Regression
(Baseline)
SVM RBF
(Main Model)
MLP
(Flexible)


Overfitting Analysis — Train vs Test Accuracy Gap
5% threshold
Train - Test Gap
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0.001
0.022
0.034
Logistic Regression
(Baseline)
SVM RBF
(Main Model)
MLP
(Flexible)

3.3. Hypothesis Testing

3.3.1. Primary Hypothesis [Laiba Khan]

The primary hypothesis stated:

" A model that is strictly linear will not work if the clothing items get confused on pairs of clothes, moderate nonlinear complexity is needed, and very flexible models will over fit.

Three pieces of evidence support this:

- Ceiling of the linear model: Logistic Regression achieved a maximum test accuracy of 81.81%. It was trained for 1000 iterations until convergence, but it could not get any better. This is its basic limited capacity in a linear model. The values of F1 for the shirt (0.56) and pullover (0.68) confirm that it fails systematically on pairs of confusable items just as predicted.
- Nonlinear improvement: SVM with RBF kernel improved to 83.54% after being trained fairly using the same 48,000 samples, which resulted in a significant improvement over logistic regression, 1.73%. The nonlinear curved boundary of the RBF kernel gave extra discriminative power which was not obtained by the linear model.
- Learned representations: MLP jumped to 89.08% , a 7.27% improvement than logistic regression. The fine local texture differences between pairs of confusable characters were captured by learned nonlinear feature combination. Shirt improved from F1=0.56 to 0.73, Pullover from 0.68 to 0.81, Coat from 0.74 to 0.84.
- Confirming the overfitting warning: The subset SVM (10,000 samples, radius = 1.0) gave train accuracy 86.59% but test accuracy 80.82%, a 5.77% gap. The chunked full-dataset SVM (radius = 1.5, outer_loops = 5) reduced this to a train/test gap of 2.24% (85.78% vs 83.54%) This showed that very flexible models are prone to fitting noise, thus supporting the hypothesis.

3.3.2. Alternative Hypothesis [Laiba Khan]

The alternative hypothesis stated:

Passive classification with a linear classifier on PCA reduced dimension to ~24 dimensions could be competitive.

Modeling with Logistic Regression on 50 PCA components got an accuracy score of 81.81%. MLP on the same features achieved 89.08%, a 7.27% gap. This is too wide to be considered as linear competitive. The alternative hypothesis is definitely disproved, the problem is not linearly solvable by compression using PCA. But the alternate hypothesis had its scientific justification it challenged the need for complexity before adopting it. As was noted above, Logistic Regression baseline showed that linear boundaries do have some real signal (81.81%), but they are just plain inadequate for complete 10-class separation.

3.3.3. Comparison of models [Ayesha Zulfiqar]

Logistic Regression vs SVM

Logistic Regression vs SVM with equal 48,000 samples, SVM was better with test accuracy of 83.54% as compared to LR with 81.81%. The train/test gap of 2.2% is a good generalization of the model, as compared with the previous subset SVM with a gap of 14.9%. The accuracy was highest for the classes of footwear and Dress, and it was lowest for Shirt, as some class boundaries are not easily discriminated with kernel nonlinearity alone.

SVM vs MLP:

MLP substantially outperformed SVM (89.08% vs 83.54%, a 5.54% gap). This is due to two reasons. The combination of task-specific nonlinear features was learned by backpropagation in the case of first MLP, and by a fixed similarity measure based on RBF functions in the case of SVM. The two-layered MLP (256→128) produced hierarchical representations: 256 dimensional PCA features were learned in the first layer and the 128 dimensional combinations were learned in the second layer; while SVM does not have any hierarchical structure. The biggest improvements were for the pairs that are the most likely to be confused: Shirt (+14% F1), Pullover (+13% F1), Coat (+9% F1).

AI Disclosure Statement

Tool: Claude (Anthropic, claude.ai)

Usage:

- **Coding [Laiba Khan & Ayesha Zulfiqar]:** Claude helped to develop the Pset 2 binary logistic regression into multiclass softmax, and the Pset 5 binary SVM into a One-vs-Rest multiclass SVM. It also helped in designing the chunked on-the-fly RBF kernel computation to accommodate the memory constraint and writing the results & visualization code (confusion matrices, F1 charts, overfitting analysis).
- **Writing & Editing [Laiba Khan & Ayesha Zulfiqar]:** Claude helped to structure and write the report text for Phase 0, Phase 1, and Phase 2. All text generated by AI was then reviewed, comprehended, and recrafted in our own words and incorporated into the final report.

Impact: Claude's coding suggestions were checked, tried and compared with our knowledge of the algorithms of PS2 and PS5. The basic algorithmic ideas (gradient descent, SVM dual updates, softmax) were known before they were used. All errors or bugs in theory will be our responsibility according to course policy.

4. Phase 3 Reflection

4.1. Was your original hypothesis correct? [Ayesha Zulfiqar]

There were two parts to our starting hypothesis: the main hypothesis and the alternative hypothesis.

The first hypothesis was that a strictly linear model would not be applicable for clothing pairs that are confusable, and a moderate level of nonlinear complexity would be needed. The first hypothesis was that a strictly linear model would not work for clothing pairs that are confusable, and a moderate level of

nonlinear complexity would be needed. This hypothesis was strongly confirmed by the results. Even though Logistic Regression was trained for 1000 iterations without stopping, it could not achieve much better than 81.81% accuracy on the test data. It was as predicted, systematically failing on the confusable pairs: Shirt had F1 of only 0.56, Pullover F1 of only 0.68, and Coat F1 of only 0.74. These are the same classes that were found to be visually similar in the Phase 0 prototype pictures. SVM with RBF kernel was improved to 83.54% showing that it helped with nonlinear boundaries. The learned nonlinear representations were found to be the most discriminative and MLP increased to 89.08%. The key hypothesis was also that very flexible models are prone to overfitting. This was directly confirmed when SVM with radius 1.0 obtained an accuracy of 96.53% during training and 81.63% during test which is a shocking 14.9% difference. When the radius was increased to 1.5, this was fixed and the warning "overfitting" was correct and needed to be done.

The alternative hypothesis was that PCA reduction and linear classification could perform just as well as the whole process of SVMs already would, given that 80% of variance is explained by only 24 components. This hypothesis was rejected. The difference between logistic regression (81.81%) and MLP (89.08%) is large (7.27%) which is not considered to be linear competitive. But the alternative hypothesis was a useful scientific tool it compelled us to test the simpler solutions first and only consider the more complicated solutions in this case. Linear boundaries are of course a necessary but insufficient condition to get full separation of 10 classes, as the logistic regression baseline confirmed that real signal can be captured with 81.81% accuracy, which is well above the random guess level of 10%. To sum up, the main hypothesis was correct, the alternate hypothesis was not supported by the evidence and the overfitting warning was right on the money. The project went exactly as scientific thinking would go, making beliefs based on data, testing them using models, and revising them in light of results.

4.2. Which observations were most predictive of success? [Ayesha Zulfiqar]

Some observations made during the Phase 0 proved to be very predictive and influenced the project's success. The only observation that was the most predictive was the PCA Scatter Plot. In the 2D PCA projection of Figure 5, Trouser separates is clearly identified as a separate group whereas all other clothing classes form a large mass. This was a prediction that was very accurate in two ways. As confirmed by Trouser having F1 of 0.95, 0.97 and 0.98 for all the three models respectively, the first Trouser would be the easiest class for all models. If we had second clothing classes, then these would need to be nonlinear, as confirmed by 7.27% improvement of MLP over logistic regression using only the clothing classes. The Average Prototype Images were very predictive of the difficulty within each class. There are very few differences in the overall shapes for Shirt and T-shirt, and Coat and Pullover at the global level of detail (Figure 2). These pairs are the most-confused pairs for all of the models, as shown by the confusion matrix results. Shirt had the smallest F1 for each individual model: Logistic regression: 0.56 SVM: 0.55 MLP: 0.73 The prototype images forecasted this month in advance of any models being trained. For a baseline, Pixel Intensity Distributions set a proper expectation for the performance of the Pixel. Figure 3 demonstrated that from the per class point of view, there are foreground intensity signatures even without spatial information. Clusters of trousers at high intensities and bimodal patterns for Sandal. This predicted that a flat model based on a vector model would be able to receive some discriminative information, which logistic regression found to be 81.81% instead of near random performance. The PCA preprocessing was justified with a Pixel-wise Standard Deviation Map. It can be seen in Figure 4 that the variance of the border pixels is very low and they don't contain any information. These pixels will naturally be given a small weight by PCA. This was confirmed directly

by the success in capturing 86.3% of the variance from only 50 components in PCA preprocessing. The Class Balance Chart was a successful way to remove the need for special handling. Only a small deviation (less than 1.5%) between the different classes of Figure 1 correctly predicted that plain accuracy would be a suitable metric and that no class rebalancing techniques would be necessary. The results obtained were good and understandable from all three models after using no special treatment in either plain accuracy or macro F1.

4.3. Which observations were misleading? [Laiba Khan]

Most of the Phase 0 observations were informative, but some were misinformed or exaggerated.

The PCA Scatter indicated closer separation than was actually observed. The 2D PCA scatter indicated that the cluster of the Trouser was clearly separate and indicated partial separation between the ankle boot and Bag. This gave the impression that there might be some kind of clustering structure out there that nonlinear models can utilize. When the same image set is used for training, the SVM, specifically designed for such geometry, could only be improved by 1.73% over logistic regression. The PCA scatter is a linear 2D projection of 784 dimensional data in only 2 of 50 principal components. It exhibited some structure but overstated the amount of cleanness in the models' exploitation of it. The overlap among the clothing classes was real, but the seemingly unique nature of some footwear classes was due in part to the particular 2D projection used.

Foreground Intensity Signatures exaggerated the value of global statistics. After removing the background pixels, intensity patterns were apparent in Figure 3: Trouser at high intensities, Sandal bimodal, Bag flat and broad. This indicated that there is indeed a discriminative signal in intensity statistics on a global scale. But when the total brightness of each pixel was examined in Non-Illustrative Figure 5, there was a huge amount of overlap between the classes. The differences in intensity of each class are real but slight enough to account for the logistic regression stopping at 81.81% but not for that it was 81.81%. While the observation was right, there was a suggestion of more discriminative power than is found in global statistics.

The Prototype Images minimized the within-class variation. In Figure 2, clean average prototype images are displayed that appear to be ideal representatives of each class. The averages are quite misleading, as they average out every noise and variation in thousands of images. Individual images of real people are considerably more noisy perhaps a T-shirt may be at a slightly different angle, crumpled or lit differently. This intra-class variation was recognised as an area of weakness in Phase 0, where intra-class variation was not directly measured. In practice this in-class noise is one of the factors that limited the accuracy of MLP to 89.08% without being higher. The prototype pictures appeared to present a somewhat rosy picture of the separability of classes.

The unreliable Non-Illustrative t-SNE suggested structure was used. Non-Illustrative Figure 3 employed an incorrect set of parameters for the t-SNE: perplexity 5 and only 500 samples. The resulting plot revealed some suggestive color groupings which appeared to be possibly meaningful clusters. This was rightly recognised as an artifact and the figure was rejected. But the real point here is that it appeared in a "cluster-like" manner, which may have tempted one to be overly confident in the class separability if we had not been careful about the parameters. As demonstrated in this lesson, the t-SNE algorithm is very sensitive to the choices of the parameters, and should not be used without careful validation of

parameters.

4.4. How would u pivot if u started over? [Laiba Khan]

With the benefit of hindsight, there are things we would do differently if we had the opportunity to go back to the beginning of the project. In Phase 0, we would measure the variation within each class. The biggest gap in our Phase 0 was the absence of any formal measurement of within-class variation. Visually, we noticed similarities between Shirt and T-shirt, but did not calculate the distance between the prototype images of each class or compute the distance from a sample to its class prototype. The cosine similarity values between all pairs of class prototype vectors would have provided a numeric measure of the difficulty of pairs of classes to confuse not just a visual measure. This would have increased the mathematical claims in the Phase 1 evidence table over the visual observations.

We would have included a PCA scatter outside of only PC1 vs PC2. The first two principal components were used in our Phase 0 PCA analysis. Projections between pairs of PCs such as PC1 vs PC3 and PC2 vs PC3 may have shown other separability structure or perhaps the separability of clothing remained unchanged even after changing the direction of the projection. This would have made our claim that linear projections are not sufficient to separate the clothing classes stricter and more definite than relying on one 2D view. We would select KNN model rather than the SVM model for a fair chance. SVM was not the best main model for this dataset and situation; having been tested on Phase 2, SVM has learned from its mistakes to improve its performance on the test data. Having already tested SVM on Phase 2, it was not the best main model for this dataset and situation. However, because of this basic restriction of the memory requirement of kernel SVMs, $O(m^2)$ where m is 48,000, SVM cannot be applied to the entirety of 48,000 samples without approximations or workarounds. We bypassed this by writing on-the-fly float32 kernel computation which came with a speed cost, and we had to have fewer outer_loops which made the SVM perhaps under-optimized.

The nonlinear model KNN would have been more appropriate; there would have been no need to train the model, the model would scale to the entire dataset, and the model would be justified by the cluster structure of the PCA scatter plot, and it could be very easily implemented from scratch with only numpy distance computations. The comparison with logistic regression and MLP would have been fairer, as all these models would have been provided the same preprocessing, the same number of 48,000 samples, and no memory workarounds.

A proper three-way split would be used for the SVM hyperparameter selection. We split the 10,000 sample into training and validation sets, and then used a 2,000 sample to validate our radius search. This is equivalent to choosing the radius from just 8 000 training samples, and testing on just 2 000 samples a really small search. The preferred way would be to reserve a validation set of 48,000 samples for the hyper parameter selection, which is neither used for training nor used for testing. This would provide more accurate estimates of radius.

We would experiment with a few random seeds for train-test split. For train-test split we picked one fixed seed of 42. However, in principle our reported accuracies are a function of which out of 12,000 samples were selected for the test set. Multiple seeds would be useful to report the mean and variance of accuracy, and provide a more accurate estimate of model performance with the correct uncertainty bounds. If applicable, mean and variance would be reported across folds, which is not possible with only a single fold. We would brainstorm feature engineering ideas before arriving at models. Instead of going

directly from PCA components to model training we could have engineered features from the PCA-reduced images. Spatial center of mass of bright pixels, simple edge detection statistics, etc. for example to compute the ratio of foreground to background pixels per image. The handcrafted features could have a much more substantial impact on the performance of logistic regression and provided insights on what information the PCA components are capturing. In the data visualization pdf they talked about feature engineering, but we skipped that and went straight to from PCA to modelling. The cross-validations would be done properly. We chose to not use k-fold cross validation due to memory requirements of SVM. Logistic regression and MLP would have been feasible and performance estimates with confidence intervals would have been more reliable. The 5-fold cross validation would involve 48,000 training samples, and 9,600 validation samples per fold, rather than our one sample at a time approach for validation.

We would have a little more of a head start. We used a softmax logistic regression on PCA features as our baseline. Logistic regression with L2 regularization. Ridge regression style would be a stronger baseline that is still linear. Linear models can usually be regularized, which can help them to generalize better, particularly if the features are correlated as PCA components are. If it was found that regularized logistic regression performed significantly better than logistic regression then it would be indicative of instability in the linear model, not capacity limits.

Statement of Contribution

- **Laiba Khan (2023-EE-102):** Exploratory Data Analysis, Illustrative Figures 1, 2, 3 (Phase 0); Problem statement and class balance analysis (Section 2.1.1); Spatial structure and visual group separation (Section 2.1.2); Main hypothesis formulation (Section 2.2.1); Model selection table contributions; Baseline Logistic Regression implementation including multiclass softmax extension, numerical stability fix, and gradient descent optimizer (Section 3.1.1); MLP implementation including architecture selection, early stopping, and hyperparameter tuning (Section 3.1.3); Implementation challenges for Logistic Regression and MLP (Section 3.1.4); Hypothesis testing , primary and alternative hypothesis analysis (Sections 3.3.1 & 3.3.2); Reflection on misleading observations (Section 4.3); Discussion of how the project would be approached differently if restarted (Section 4.4).
- **Ayesha Zulfiqar (2023-EE-101):** Exploratory Data Analysis, Illustrative Figures 4, 5 and Non-Illustrative Figures 1, 2, 3 (Phase 0); Pixel intensity and foreground signal analysis (Section 2.1.3); Spatial redundancy and linear separability analysis (Section 2.1.4); Alternative hypothesis formulation (Section 2.2.2); SVM with RBF Kernel implementation including on-the-fly memory-efficient kernel computation, float32 optimization, and radius selection (Section 3.1.2); SVM implementation challenges (Section 3.1.4); Model comparison: Logistic Regression vs SVM vs MLP (Section 3.3.3); Reflection on correctness of original hypothesis (Section 4.1); Discussion of which Phase 0 observations were most predictive of success (Section 4.2).